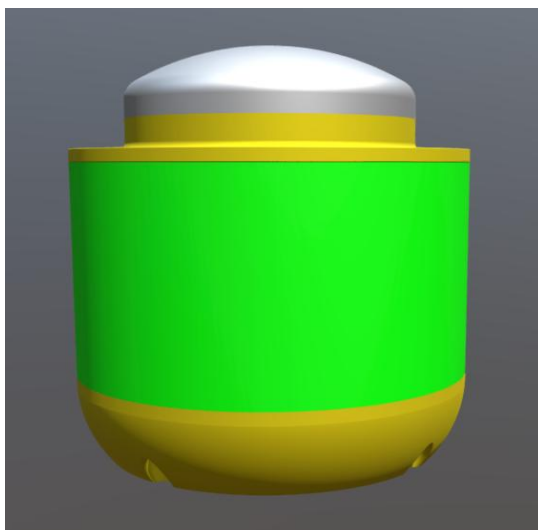


SurVN RTK

Part 1: Introduction

- **SurVN RTK** is firmware that runs on **ESP32**, developed to create an **RTK** solution with **Tilt Compensation** support for the **UM981/UM981S** module at a reasonable cost while still meeting surveying needs well, and is also compatible with many survey software programs and apps that support **NMEA** data.
- The ESP32 acts as a smart bridge between the GNSS receiver and the mobile device. The firmware provides a Web interface to configure the receiver, process tilt-compensation data, and forward NMEA data. At the same time, the ESP32 receives RTCM correction data and sends it via **Bluetooth (SPP)** to the GNSS receiver.



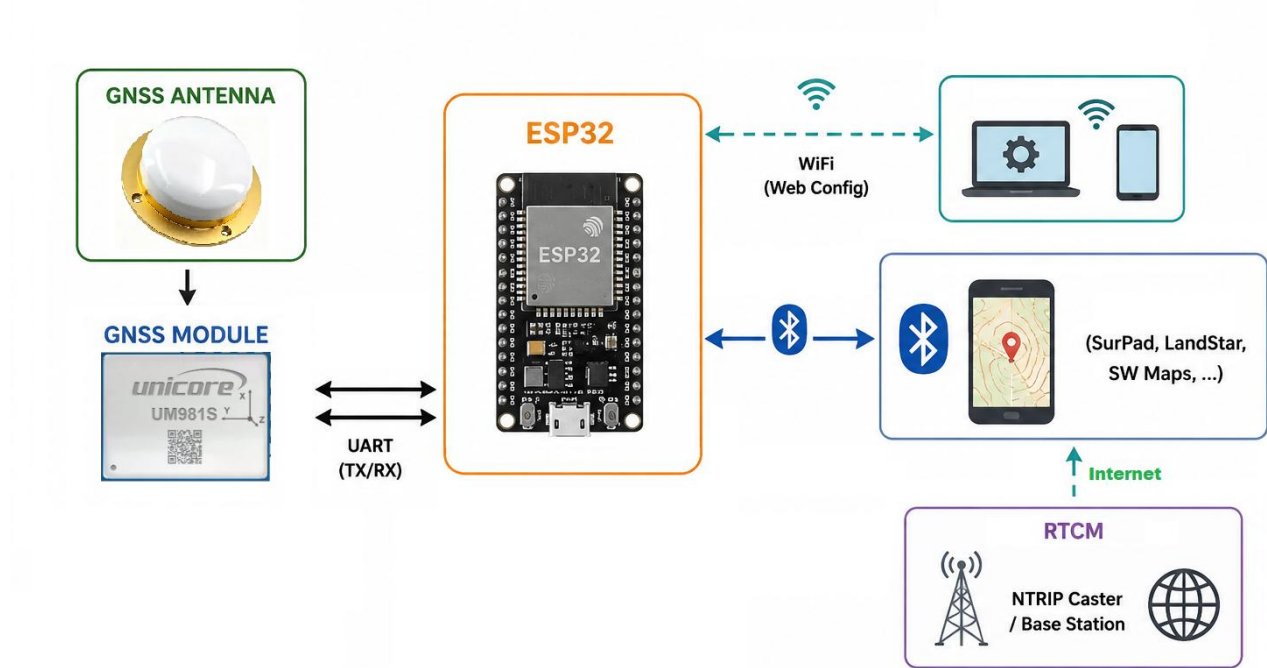
❖ Advantages

- **Low cost**, can be self-assembled and upgraded.
- **Fully automatic first-time UM981/UM981S configuration through the Web interface**, no need to send manual GNSS configuration commands.
- **Automatically saves configuration**, ready to use for subsequent power-ups.
- **Supports OTA firmware updates over Wi-Fi**, no USB cable connection required.
- **Compatible with many surveying apps** such as SurPad, LandStar, SW Maps, etc.
- **Stable Bluetooth connection**, ensuring continuous NMEA transmission and RTCM reception.

❖ Performance

- Achieves fast RTK FIX initialization time.
- RTK accuracy as published by the GNSS module manufacturer. ([see more](#))
- The accuracy stated above is a specification published by the GNSS module manufacturer and is for reference only. Actual performance depends on environmental conditions, antenna quality, RTCM source, and the user's usage process.

Part 2: System overview



Part 3. Project resources [\(Google Drive\)](#)

All project resources are stored on Google Drive and are always updated to the latest version, including:

- *Parts list (Parts_List).*
- *3D-printed case design files (STL).*
- *Firmware **SurVN_RTK_V1.1.bin***
- *Flash Download Tool*
- *OTA firmware update folder (**Update**)*

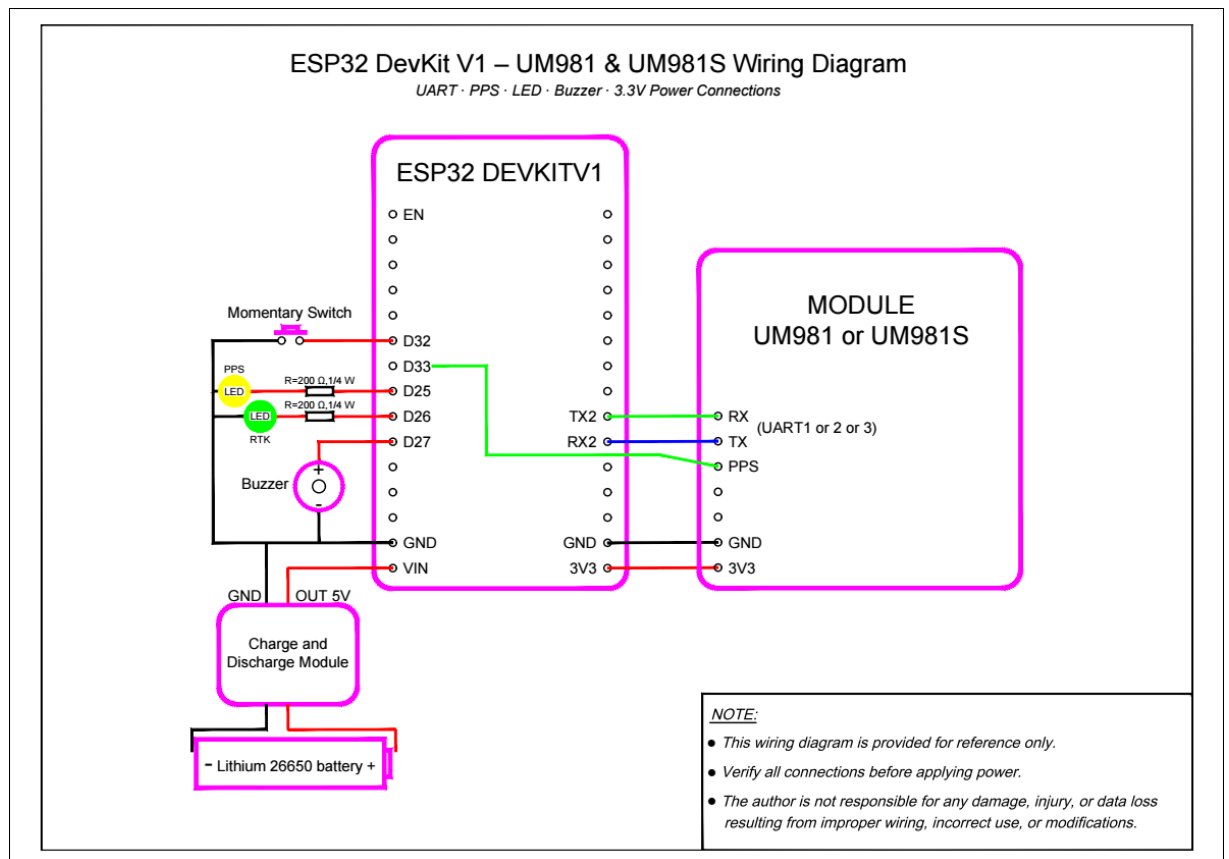
❖ 3D printing material recommendations

- The firmware **SurVN RTK** was designed and tested using 3D-printed **PETG** plastic, as this material has good mechanical strength, higher heat resistance than common plastics, and warps less during printing — making it suitable for building an enclosure for outdoor use.
- To ensure the geometric accuracy of the device after assembly, especially the mounting positions of the **GNSS module, antenna, IMU**, and other parts, the case components must be printed at the exact design dimensions. Therefore, the **material shrinkage setting(Shrinkage)** on the 3D printer must be calibrated before printing the final parts. ([refer to BambuLab's documentation](#))

Part 4: Assembly

Refer to this [video](#) for a similar assembly on the [SurveyorVN](#)

Part 5: Wiring diagram



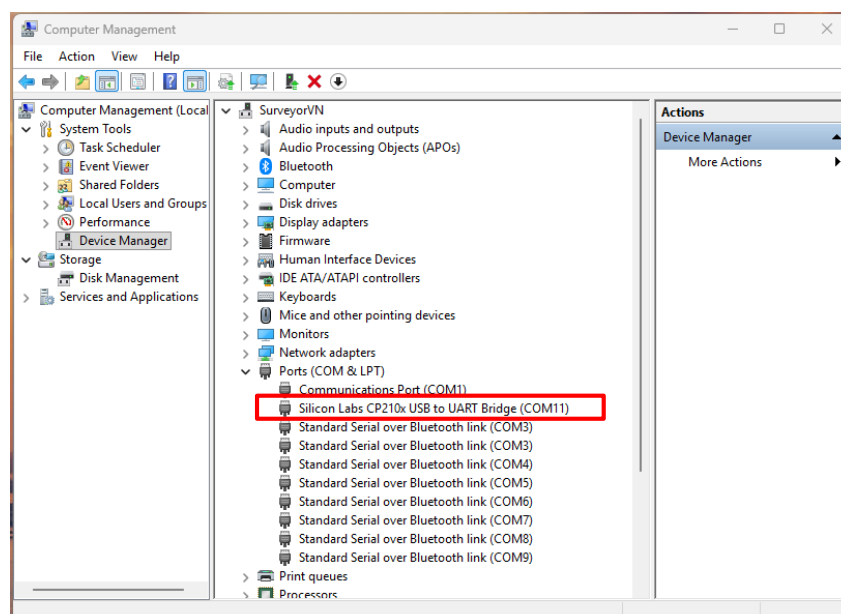
Part 6: Flashing the firmware with Flash Download Tool

❖ Preparation:

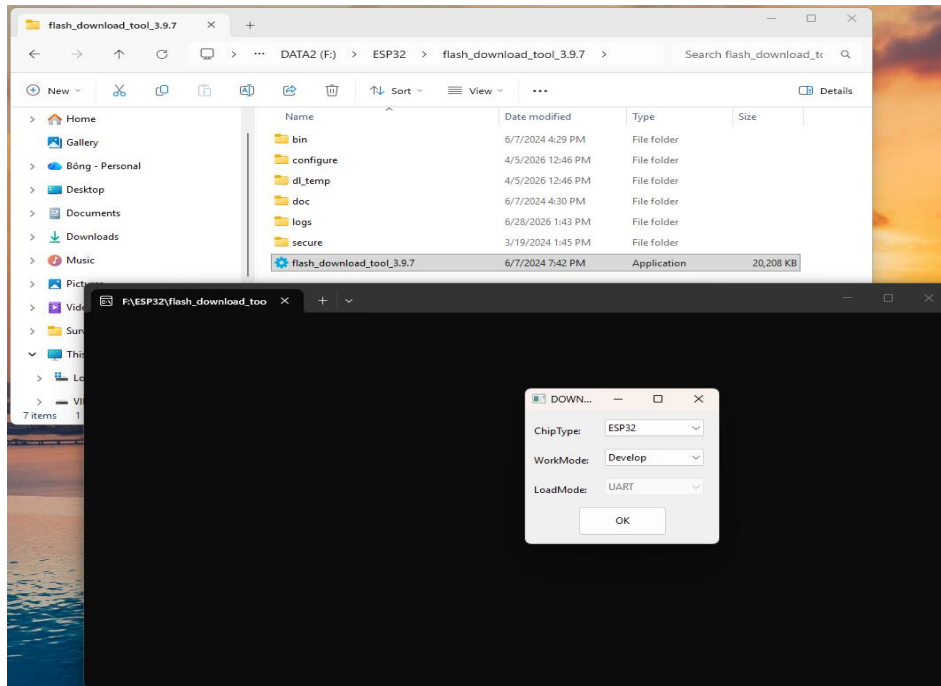
- The **Flash Download Tool** software
- Firmware **SurVN_RTK_V1.1.bin** file
- A cable to connect the ESP32 to the computer (PC)
- Install the USB-Serial driver if not already installed (CP210x or CH340 depending on the board)

❖ Flashing steps

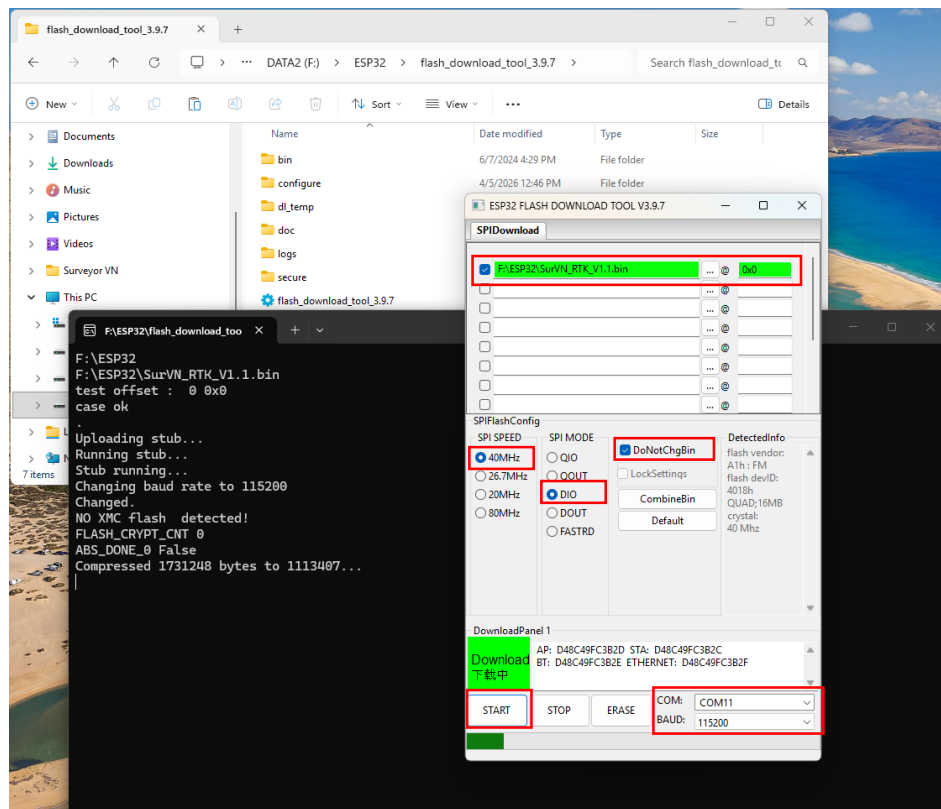
1. Plug the connection cable between the **ESP32** and the **PC** via USB cable.



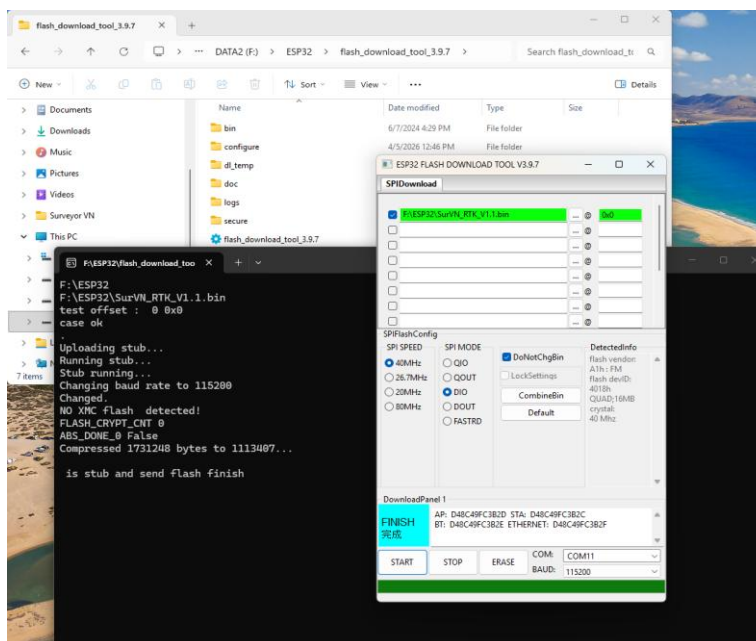
2. Confirm whether the ESP32's **COM** port has been detected (here my ESP32 is detected as COM11)



3. Open the **Flash download tool** program, and select the settings shown above: **ESP32 – Develop – UART → OK**



4. Find and select the file **SurVN_RTK_V1.1.bin** at address **0x0** as shown above → Press **START** in Flash Download Tool (if the program does not run automatically, hold down the **BOOT** button on the ESP32) and wait for the progress bar to run as shown above, then release the **BOOT** button.



- Once flashing is complete, press the **EN (Reset)** button, or unplug and reconnect the USB cable to restart the ESP32.

❖ Post-flash check

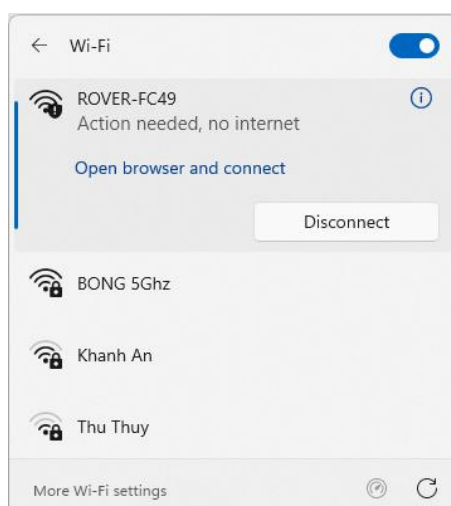
- After restarting, the **ESP32** will broadcast Bluetooth. To switch to Wi-Fi, hold the Tilt button for >2s; a sound will confirm the switch.
- The ESP32 broadcasts a Bluetooth and Wi-Fi AP named **ROVER-XXXX**, (XXXX is a unique code for each device)
- Connect to the Wi-Fi (*choose to use it without internet*) → open a Web browser → go to **192.168.4.1** and the SurVN RTK Webinterface appears → the installation was successful.

❖ How the Tilt button works

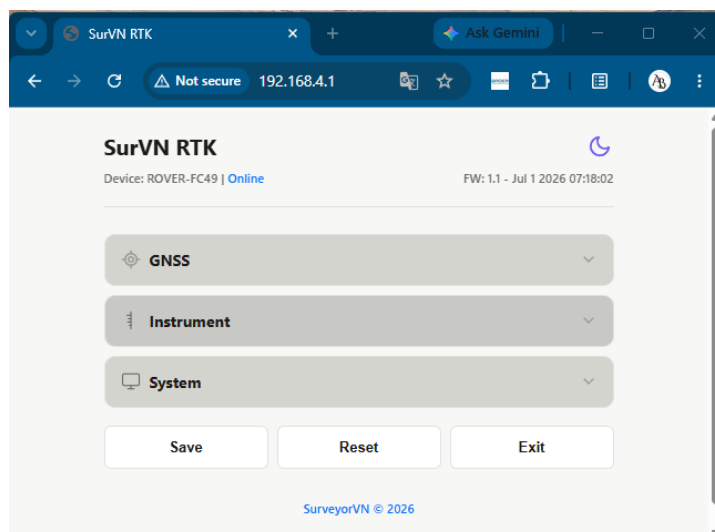
- Press and release once: turns off tilt compensation; press and release a second time to turn tilt compensation back on
- Hold for >2s (release when you hear the sound): Switches back and forth between Wi-Fi and Bluetooth

Part 7: Initial configuration

- Use a mobile phone or a PC to connect to the device's Wi-Fi AP.



- Go to the configuration web page (192.168.4.1)



The first-time web setup includes the following main parameters:

- **GNSS**
 - Measurement rate 2Hz (equivalent to 2 position updates per second) or 5Hz (equivalent to 5 position updates per second)
 - Satellite systemson/off
- **Tool**
 - the distance from the base of the device to the phase center of the GNSS antenna (Phase center offset)
 - Pole length (Pole Length)
 - ***Height entered into the survey software = Pole length + the distance from the base of the device to the phase center of the GNSS antenna.***
- **System**
 - The IMU offset value relative to the GNSS antenna phase center must be measured accurately to the millimeter along the coordinate axes marked on the module. ([refer to Unicore's documentation](#))



- The IMU offset value (X, Y, Z) — important for tilt compensation (after entering the **Offset** parameters, you must press **the Save button** to apply them).

- After pressing **Save**, select **Factory reset** to let the program set up the RTK configuration for the module for the first time. A successful factory reset will show a notification message as shown below; the program will automatically restart and enter Bluetooth mode, and you can then proceed with IMU calibration.

The screenshot shows a web interface with two main sections. The top section, titled "EMERGENCY RECOVERY", contains a button with a warning icon and the text "Factory reset". Below this button is a green message: "Factory reset completed. Device reconfigured." The bottom section, titled "FIRMWARE UPDATE", features a "Choose File" button next to the text "No file chosen", and an "Upload & update" button with an upward arrow icon. At the very bottom of the interface are three buttons: "Save", "Reset", and "Exit".

Part 8: First-time IMU calibration

- Take the **RTK** outside under an open sky, use the **SW Maps app** to connect to the **RTK** via the Bluetooth name **ROVER-XXXX**, and wait for the **RTK** unit to receive the satellite signal (**NMEA**). At that point, in the **SW Maps app**'s device connection section, you will see the antenna height set to the value entered in the SurVN RTK Web— confirming that a signal has been received.

The screenshot shows the "Instrument" configuration page in the SW Maps app. It has a title bar with a back arrow and the word "Instrument". Below the title bar, there are two input fields. The first is labeled "PHASE CENTER OFFSET" and contains the value "80" with the unit "mm" on the right. Below this field is the text "Fixed constant for this device type". The second input field is labeled "POLE LENGTH" and contains the value "1.700" with the unit "m" on the right. At the bottom of the page, there is a green box containing the text "Survey app antenna height" and the value "1.780 m".

*Example: the antenna height value for this unit each time it restarts is **1.780m***

- You need to connect to Ntrip in the **SW Maps app** for the device to get an **RTK Fix**. Once that happens, you will see the tilt-compensation icon in the app; follow the on-screen instructions until calibration reaches 100% (*the device will emit a single long beep*), indicating that calibration was successful.
- Once the IMU has been successfully calibrated, on subsequent power-ups you only need to achieve RTK Fix and then gently tilt the pole **left - right - forward - backward** to activate tilt-compensation mode.

Part 9: Common troubleshooting

Symptom	Possible cause	Solution
No Bluetooth or Wi-Fi name shown: ROVER-XXXX	ESP32 has not been restarted; not powered; firmware not flashed; ESP32 faulty	Reflash the firmware
Cannot connect to Bluetooth	Bluetooth is being held by another phone's connection	Turn Bluetooth off/on or restart the ESP32
Web shows Device: ROVER-XXXX Offline	Not connected to the ESP32's Wi-Fi	Check whether the ESP32 is currently broadcasting Wi-Fi or not.
GNSS icon not blinking	Incorrect UART wiring	Check Tx2(ESP32) – Rx(GNSS) and Rx2(ESP32) – Tx(GNSS)
No RTK Fix	Missing NTRIP / weak signal	Check the correction source
Inaccurate tilt	Incorrect IMU offset	Recalibrate the XYZ offset
Web page does not load	Wrong IP address	Try the IP again: 192.168.4.1

Part 10: Contact information & support

- Zalo: 0985.629.650
- Gmail: bongdvtd@gmail.com
- youtube.com/@SurveyorVN
- facebook.com/SurveyorVN/
- tiktok.com/@surveyor.vn

Terms of use: The SurVN RTK firmware is free to use for **300** device power-ups. To use it long-term and receive updates, please contact the **author** to activate a permanent license for your device.

Disclaimer:

- The firmware is provided as-is, without any warranty.
- The user is solely responsible for the assembly, configuration, and use of the device.
- The author is not responsible for any hardware damage, data loss, or measurement errors resulting from improper use or assembly.
- The firmware may not be copied, modified, or distributed without the author's consent.